

22. The study of ageing is known as –

- (a) Gerontology (b) Ethnology
(c) Anthropology (d) Thanatology

U.P.P.C.S. (Mains) 2014

Ans. (a)

See the explanation of above question.

23. Pedology is the science related to the study of :

- (a) Atmosphere (b) Soil
(c) Pollutants (d) Seeds

R.A.S./R.T.S. (Pre) 1999

Ans. (b)

Pedology deals with the study of soil formation, soil morphology, and soil classification.

24. Study of bones, are under which branch of science?

- (a) Osteology (b) Orology
(c) Serology (d) Geology

U.P.P.C.S. (Pre) 1990

Ans. (a)

Osteology	–	Study of bones
Orology	–	Study of mountains
Serology	–	Study of serum
Geology	–	Study of internal structure of the earth and matter found inside it

25. The study of bones is done under the branch of science called

- (a) Geology (b) Serology
(c) Orology (d) Osteology
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

See the explanation of above question.

26. The study of snakes is known as :

- (a) Serpentology (b) Ornithology
(c) Herpetology (d) Ichthyology

Uttarakhand P.C.S. (Mains) 2002

Ans. (a)

Serpentology is a branch of Zoology, which deals with the study of snakes. Ornithology is a branch of Zoology that deals with the study of birds. Herpetology is the branch of Zoology related to the study of amphibians (including frogs, toads, salamanders, newts and Gymnophiona) and reptiles (including snakes, lizards, turtles, terrapins, tortoises, crocodilians and the tuataras). Ichthyology also known as fish science, is the branch of biology devoted to the study of fish.

27. Find out the odd one out from the following :

- (a) Physiology (b) Psychology
(c) Pathology (d) Bacteriology

Uttarakhand U.D.A./L.D.A. (Pre) 2007

Ans. (b)

Physiology, Pathology, and Bacteriology are the branches of biology, whereas Psychology is the study of mind and behaviour and is different from the other three.

28. What is hydroponics?

- (a) Growing plant without soil
(b) Grafting plant
(c) Study of vegetables
(d) Soil conservation

U.P.P.C.S. (Mains) 2009

Ans. (a)

Hydroponics is a subset of hydroculture and is a method of growing plant using mineral nutrients solutions in water without soil.

Evolution of Life

Notes

- Earth formed around 4.54 billion years ago, approximately at one-third of the age of universe by accretion from the solar nebula.
- Several theories/postulations have been propounded in reference to the origin of life on Earth, but most accepted theory among them is the 'Oparin-Haldane Theory'.

Oparin-Haldane Theory of Origin of Life :

- According to this theory life originated on early Earth through physico-chemical processes of atoms combining to form molecules, which in turn reacting to produce inorganic and organic compounds. Organic compounds interacted to produce all types of macromolecules which organized to form the first living system or cells.
- Thus, according to this theory 'life' originated upon our Earth spontaneously from non-living matter. First inorganic compounds and then organic compounds were formed in accordance with ever-changing environmental conditions. This is called chemical evolution which cannot occur under present environmental conditions upon Earth. Conditions suitable for origin of life existed only upon primitive Earth.

- Hypothesis given by Oparin about origin of life is also known as Materialistic Theory which was published in his book '**Origin of Life**'.
- Modern views regarding the origin of life include both chemical evolution and biological or organic evolution.

A. Chemical Evolution (Chemogeny) :

1. The Atomic Phase :

- Early Earth had innumerable atoms of all those elements (e.g. hydrogen, oxygen, carbon, nitrogen, sulphur, phosphorus) which are essential for the formation of protoplasm. Atoms were segregated in three concentric masses according to their weight -
- (a) The heaviest atoms of iron, nickel, copper, etc. were collected in the centre of the Earth.
 - (b) Medium weight atoms of sodium, potassium, silicon, magnesium, aluminium, phosphorus, chlorine, fluorine, sulphur etc. were collected in the core of the Earth.
 - (c) The lighter atoms of nitrogen, hydrogen, oxygen, carbon etc. formed the primitive atmosphere.

2. Formation of Inorganic Molecules :

- Free atoms combined to form inorganic molecules such as H_2 (hydrogen), N_2 (nitrogen), H_2O (water vapour), CH_4 (methane), NH_3 (ammonia), CO_2 (carbon dioxide), etc. Hydrogen atoms were most numerous and most reactive in primitive atmosphere.
- Hydrogen atoms also combined with nitrogen forming ammonia (NH_3). So the water and ammonia were probably the first inorganic molecules of primitive Earth.

3. Formation of Simple Organic Molecules (Monomers) :

- The early inorganic molecules interacted and produced simple organic molecules such as simple sugars (e.g. ribose, deoxyribose, glucose etc.), nitrogenous bases (e.g. purines, pyrimidines), amino acids, glycerol, fatty acids etc.
- Terrestrial rains must have fallen. As the water rushed down, it must have dissolved away and carried with it salts and minerals, and ultimately accumulated in the form of oceans. Thus ancient oceanic water contained large amounts of dissolved NH_3 , CH_4 , HCN , nitriles, carbides as well as various gases and elements.
- Some external sources must have been acting on the mixture for reactions. These external sources might be (i) solar radiations such as ultraviolet rays (UV rays), X-rays etc., (ii) energy from electrical discharges like lightning, (iii) high energy radiations are other sources of energies (probably unstable isotopes on primitive Earth).

There was no Ozone layer at that time in the atmosphere.

- A soup like broth of chemicals formed in oceans of the early Earth, from which living cells are believed to have appeared, was termed by Haldane as **prebiotic soup**, also called hot dilute soup. Thus the stage was set for combination of various chemical elements. Once formed, the organic molecules accumulated in water, because their degradation was extremely slow in absence of any life or enzyme catalysts.
- The formed molecules accumulated in ocean bounded by a covering forming coacervates which have the capacity to self-replication.

Experimental evidence for Abiogenetic Molecular Evolution of Life :

- **Stanley Miller** in 1953, who was then a graduate student of **Harold Urey** at the University of Chicago, demonstrated it clearly that ultraviolet radiation or electrical discharges or heat or combination of these can produce complex organic compounds from a mixture of methane, ammonia, water and hydrogen. The ratio of methane, ammonia and hydrogen in Miller's experiment was 2 : 1 : 2.
- Miller circulated four gases : methane, ammonia, hydrogen and water vapour in an airtight apparatus and passed electrical discharges in it from electrodes at $800^\circ C$. Then he passed the mixture through a condenser.
- He circulated the gases continuously in this way for one week and then analysed the chemical composition of the liquid inside the apparatus. He found a large number of simple organic compounds including some amino acids such as alanine, glycine and aspartic acid.
- Miller conducted the experiment to test the idea that organic molecules could be synthesized in a reducing environment.
- It is considered that the essential building blocks such as nucleotides, amino acids etc. of living organisms could thus have formed on the primitive Earth.
- Early life on Earth formed around 4.0 billion years ago.
- Membrane bound prokaryotes originated around 3.8 billion years ago, which were previously **heterotrophs** and later on became **autotrophs**. Autotrophs were just like the blue-green algae of modern period.
- Approximately 2.0 billion years ago eukaryotic cells were formed, from which all the organisms (except bacteria) of our planet evolved.

B. Biological or Organic Evolution :

Fundamental presumptions and principle of organic Evolution :

- The fundamental presumption of organic evolution is – ‘the modification of living organisms during their descent, generation by generation from common ancestors’.
- According to the concept of organic evolution, the present day animals and plants have been evolved by a process of gradual change in the earlier simple forms of life, which took place in millions of years.
- The theories of organic evolution explains convincingly the origin of life. It also explains how the wide variety of plants and animals came into existence in the world.
- In organic evolution, there are different theories of evolution, but following theories are accepted universally. They are :
 - (i) Theory of Inheritance of Acquired Character (Lamarckism)
 - (ii) Theory of Natural Selection (Darwinism)
 - (iii) Mutation Theory (Hugo de Vries)

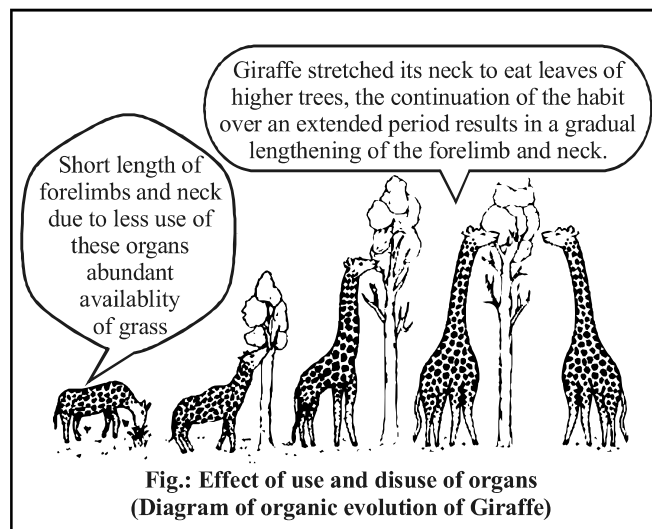
I. Theory of Inheritance of Acquired Character (Lamarckism) :

- Lamarckism is the first theory of organic evolution, named after Jean-Baptiste Lamarck (1744-1829), a French biologist.
- Lamarck's famous book is ‘**Philosophie Zoologique**’ (1809).
- He announced in 1801, a theory of organic evolution which has been known as ‘Theory of Inheritance of Acquired Character’.
- His evolutionary ideas are—
 - Internal forces of life tend to increase the size of the organism and because of an ‘**inner want**’ new structures appear.
 - The direct environmental effect over living organisms.
 - Use or disuse of organs.
 - Inheritance of acquired character.

Examples -

- Long neck of giraffe due to lack of surface vegetation gradually.
- Aquatic birds - they had to go to water due to lack of food etc. Some structures ‘web’ between their toes developed and wings for fly gradually reduced.
- Flat fishes (deep sea fishes).
- Whales lost their hind limbs.

- The wading birds (e.g. Jacana) developed its long legs through generation of sustained stretching to keep the body above the water level.



II. The Theory of Natural Selection (Darwinism) :

- Charles Darwin explained natural selection in his book ‘**On the Origin of Species**’ (1859 ; Full Title - On the Origin of Species by Means of Natural Selection, or the Preservation of Favoured Races in the Struggle for Life). It is considered to be the foundation of evolutionary biology.
- It includes the following elements -
 - The universal occurrence of variation
 - Over production (rapid multiplication)
 - **The struggle for existence** : Intra-specific struggle between same species. Inter-specific struggle between different species. Environmental struggle due to earthquake, tidal waves, the burning of volcanoes are all causes for killing large populations.
 - **Survival of the fittest (Natural Selection):** Organisms struggle for existence and organisms with advantageous characters survive, while those which lack such variations perish. Thus, individuals having favorable variations have better chances of living long enough to reproduce.
 - **Inheritance of useful variations** : The organisms after getting fitted to the surroundings transmit their useful variations to the next generation, while non-useful variations are eliminated.
- **Neo-Darwinism** : Neo-Darwinism is a modified form of Darwinism. Neo-Darwinism term generally used for describing any integration of Darwin’s theory of evolution by natural selection with Gregor Mendel’s

theory of genetics. George Romanes first used this word in 1895.

- **Modern synthetic theory** is the result of work of a number of scientists namely **T-Dobzhansky, R.A. Fisher, J.B.S. Haldan, Small Wright, Ernst Mayr** and **G.I. Stebbins**. Stebbins in his book '**Process of organic evolution**' discussed the '**synthetic theory**'. This is one of the proven theories of organic evolution. It includes the following factors–
1. Mutation
 2. Variation or Recombination
 3. Heredity
 4. Natural Selection
 5. Isolation

III. Mutation Theory :

- Dutch botanist **Hugo de Vries**, in his book '**The Mutation Theory**' (1901) proposed this theory.
- He observed on 'Evening Primrose', **Oenothera lamarckiana**. He studied this plant in wild forms for many years continuously and observed certain spontaneous changes in some of these wild plants.
- Mutation Theory states that evolution is a jerky process where new varieties and species are formed by mutations (discontinuous variations) that function as raw material of evolution.

Salient features of mutation theory are :

- Mutation appear all of a sudden. They become operational immediately.
- The same type of mutation can appear in a number of individuals of a species.
- All mutations are inheritable.
- Useful mutations are selected by nature. Lethal mutations are eliminated. However, useless and less harmful ones can persist in the progeny.
- Accumulation of variations produce new species, sometimes a new species is produced from a single mutation.

Important Facts :

- **Archaeopterix** : It looks like a bird. It has wings and beak like birds. However its teeth and tail are like those of reptiles.
- Archaeopterix is, therefore, considered as a connecting link between reptiles and birds, thereby suggesting that birds have evolved from reptiles.
- The fossils of Archaeopterix was first discovered from limestone deposits near Solnhofen, Germany.

- Archaeopterix lived in the late jurassic period of Mesozoic era around 150 millions years ago.
- **Dinosaurs** : During the **Mesozoic** or 'Middle Life Era', life diversified rapidly and giant reptiles dinosaurs and monstrous beasts roamed the earth. The period, which spans from 252 million years ago to about 66 millions years ago was also known as the age of reptiles or the age of dinosaurs.
- Mesozoic era has been divided into three periods viz– the Triassic (251-199.6 millions years ago), the Jurassic (199.6 to 145.5 million years ago) and the Cretaceous (145.5 to 65.5 millions years ago).
- **Cro-Magnon** : The earliest known Cro-Magnon remains are between 35000 and 45000 years old based on radiometric dating.
- Cro-Magnon had powerful bodies, which were usually heavy and solid with strong muscles with height about 180 cm.
- The Cro-Magnon had straight forehead like modern humans. Their faces were short and wide with a large chin. Their brains were slightly larger than the average human's of today. The brain capacity of Cro-Magnon was 1600 cc.
- The name 'Cro-Magnon' was coined by Louis Lartet, who discovered the first Cro-Magnon skull in Southwestern France in 1868.
- The Cro-Magnon are considered the last direct ancestor of modern man and it was a sub-species of modern human named **Homo sapiens**.

Question Bank

1. Choose the correct chronological order of given Epochs, oldest one will come first :
 - (a) Holocene > Pleistocene > Pliocene > Miocene > Eocene
 - (b) Eocene > Miocene > Pliocene > Pleistocene > Holocene
 - (c) Eocene > Miocene > Pleistocene > Pliocene > Holocene
 - (d) Holocene > Pliocene > Pleistocene > Miocene > Eocene

Chhattisgarh P.C.S. (Pre) 2021

Ans. (b)

The chronological order of Epochs (from oldest to newest) is as follows :
 Paleocene > Eocene > Oligocene > Miocene > Pliocene > Pleistocene > Holocene

2. Which of the following gases was absent in the atmosphere of the primitive Earth?

- (a) Methane (b) Ammonia
(c) Oxygen (d) Carbon dioxide

U.P. R.O./A.R.O. (Mains) 2016

Ans. (c)

Primitive Earth's atmosphere was rich in methane, ammonia, water vapour, carbon dioxide, carbon monoxide, nitrogen, hydrogen, neon etc. It was a reducing atmosphere, with no free oxygen. It is likely that hundreds of millions of years separated the first biological production of oxygen by unicellular organisms and its eventual accumulation in the atmosphere.

3. Consider the following statements :

1. The Earth's magnetic field has reversed every few hundred thousand years.
2. When the Earth was created more than 4000 million years ago, there was 54% oxygen and no carbon dioxide.
3. When living organisms originated, they modified the early atmosphere of the Earth.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2018

Ans. (c)

Over the history of the Earth, its magnetic field has reversed every few hundred thousand years. Over the last 5 million years, the Earth's magnetic field reversed itself at least 20 times. Over the last 20 million years, the reversal happens about every 200 thousand to 300 thousand years. However, it has been about 780 thousand years since the last reversal. Hence, statement 1 is correct.

When Earth was created there was no oxygen in the atmosphere. Oxygen, although always present in compounds in Earth's interior, atmosphere, and oceans, did not begin to accumulate in the atmosphere as oxygen gas (O_2). The earliest atmosphere contained carbon dioxide, methane, water vapour, ammonia, nitrogen, hydrogen, etc. Hence, statement 2 is incorrect.

When living organisms originated, they modified the early atmosphere of the Earth. However, it is likely that hundreds of millions of years separated the first biological production of oxygen by unicellular organisms (like cyanobacteria) and its eventual accumulation in the atmosphere. Thus, statement 3 is correct.

4. According to recent proof, the life on Earth emerged approximately -

- (a) Before 20,000 years
(b) Before 2,00,000 years
(c) Before 20,00000 years
(d) Before 2,000,000,000 years

40th B.P.S.C. (Pre) 1995

Ans. (d)

The age of the Earth is about 4.6 billion years. This dating is based on evidence from radiometric age dating of meteorite materials and is consistent with the radiometric ages of the oldest-known terrestrial and lunar samples. According to recent proof, the life on Earth emerged about 2,000,000,000 years ago.

5. Which is the oldest organism on Earth ?

- (a) Blue-green algae (b) Fungi
(c) Amoeba (d) Euglena

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (a)

Blue-green algae is the oldest organism on Earth. They obtain their energy through photosynthesis. They are also known as Cyanobacteria. By producing gaseous oxygen as a by-product of photosynthesis, they played a greater role in dramatically changing life forms on Earth by stimulating biodiversity.

6. The book, The Origin of Species was written by :

- (a) Linnaeus (b) Lamarck
(c) Mendel (d) Darwin
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

Charles Darwin explained natural selection in his book 'The Origin of Species' (1859; Full Title - On the Origin of Species by Means of Natural Selection, or the Preservation of Favoured Races in the Struggle for Life). It is considered to be the foundation of evolutionary biology.

7. Darwin's theory of natural selection is based on -

- (a) Overproduction
(b) Struggle for existence and variation
(c) Survival of the fittest
(d) All of the above

M.P.P.C.S. (Pre) 1997

Ans. (d)

Charles Robert Darwin was an English geologist and best known for his contribution to evolution theory. Darwin published his theory of evolution in his book 'On the Origin of Species' in 1859. It states that all species of organisms arise and develop through the natural selection of small, inherited variations that increase the individual's ability to compete, survive and reproduce. It is called Darwinian theory.

8. Who propounded the theory of evolution?

- (a) Spencer (b) Darwin
(c) Wallace (d) Huxley

M.P.P.C.S. (Pre) 2010

Ans. (b)

See the explanation of above question.

9. Who explained first the evolution of life?

- (a) Newton (b) Einstein
(c) Charles Darwin (d) Lamarck

43rd B.P.S.C. (Pre) 1999

Ans. (d)

French biologist Lamarck is best known for his "Theory of Inheritance of Acquired Characters", first presented in 1801. He was the first person to explain the theory of evolution to the public. In 1809, this theory was published in his book 'Philosophie Zoologique'.

10. The mutation theory of evolution was enunciated by

- (a) Huxley (b) Darwin
(c) Lamarck (d) Hugo de Vries

U.P.P.C.S. (Pre) 2005

Ans. (d)

Hugo De Vries is known for his mutation theory of evolution. According to him, new species are not formed by continuous variations but by the sudden appearance of variations, which he named as mutations. He stated that mutations are heritable and persist in successive generations.

11. Match List-I with List-II and select the correct answer using the codes given below/the lists :

List-I	List-II
A. Theory of mutation	1. Beadle and Tatum
B. Theory of evolution	2. Jacob and Monod
C. One gene one enzyme hypothesis	3. Darwin
D. Operon concept	4. De Vries

Code :

	A	B	C	D
(a)	3	4	1	2
(b)	4	3	1	2
(c)	4	3	2	1
(d)	3	4	2	1

I.A.S. (Pre) 1998

Ans. (b)

The correctly matched lists are as follows :

Theory of mutation	–	Hugo de Vries
Theory of evolution	–	Charles Darwin
One gene one enzyme hypothesis	–	Beadle and Tatum
Operon concept	–	Jacob and Monod

12. The main factor of evolution is –

- (a) Mutation
(b) Achieved quality
(c) Sexual reproduction
(d) Natural selection

39th B.P.S.C. (Pre) 1994

Ans. (d)

According to Darwin's theory of evolution, natural selection is one of the basic mechanism of evolution.

13. In the context of organic evolution, the loss of limbs in snakes is explained by the phenomenon of :

- (a) Use and disuse of organs
(b) Adaptation to living in burrows
(c) Natural selection
(d) Inheritance of acquired characters

I.A.S. (Pre) 2002

Ans. (a)

Lamarckism (or Lamarckian inheritance) is the idea that an organism can pass one characteristic that it acquired during its lifetime to its offspring (also known as inheritance of acquired character). It is named after Lamarck who incorporated following two ideas into his theory of evolution, which are considered to be generally true :

1. Use and disuse of organs – Individuals lose characteristics which they do not require (or use) and develop characteristics that are useful.
2. Inheritance of acquired traits – Individuals inherit the acquired traits of their ancestors.

In the context of organic evolution as explained by Lamarck, the loss of limbs in snakes is explained by the phenomenon of use and disuse of organs.

14. The reason of large-scale diversity among the organisms is –

- (a) Adaptation (b) Cooperation
(c) Mutation (d) Polyploidy

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

Three basic and three accessory factors have been recognized to provide genetic variability to the gene pool of a population. Amongst the basic factors, gene and chromosomal mutations have been recognized as the ultimate sources of biological changes and hence, the ultimate 'raw material' of evolution. Gene recombinations, the third basic factor then increases the variability of individual genotypes in a population.

15. Among living organisms, which one of the following is the most responsible factor for bringing about the origin of a new species ?

- (a) Isolation (b) Mutation
(c) Natural selection (d) Sexual reproduction

I.A.S. (Pre) 2002

Ans. (b)

In biology, a mutation is a permanent change of the nucleotide sequence of the genome of an organism, virus or extra-chromosomal DNA or other genetic elements. Mutations result from damage to DNA which is not repaired or to RNA genomes (typically caused by radiation or chemical mutagens), errors in the process of replication or from the insertion or deletion of segments of DNA by mobile genetic elements. Mutations play a part in both normal and abnormal biological processes including evolution, cancer and the development of the immune system including functional diversity and among living organisms, is the most responsible factor for bringing about the origin of a new species.

16. Latest ancestor of modern human is :

- (a) Java man (b) Cro-Magnon man
(c) Neanderthal man (d) Peking man

39th B.P.S.C. (Pre) 1994

Ans. (b)

Cro-Magnon is a common name that has been used to describe the first early modern humans. The earliest known remains of Cro-Magnon-like humans are radiocarbon dated to 43,000 years before. Cro-Magnons were robustly built and powerful. Their body was generally heavy and solid with a strong musculature. The face was short and wide. The chin was prominent. The brain capacity was about 1,600 cubic centimeters, larger than the average for modern humans. However, recent research suggests that the physical dimensions of so-called 'Cro-Magnon' are not sufficiently different from modern humans to warrant a separate designation.

17. The word 'Denisovan' is sometimes mentioned in media in reference to :

- (a) fossils of a kind of dinosaurs
(b) an early human species
(c) a cave system found in North-East India
(d) a geological period in the history of Indian subcontinent

I.A.S. (Pre) 2019

Ans. (b)

The **Denisovans** or **Denisova hominins** are an extinct species or subspecies of archaic human that ranged across Asia during the Lower and Middle Paleolithic. The first identification of a Denisovan individual occurred in 2010, based on mitochondrial DNA (mtDNA) extracted from a juvenile female finger bone excavated from the Siberian Denisovan's Cave in the Altai Mountains in 2008. Nuclear DNA indicates close affinities with Neanderthals. Additional specimens from Denisovan's Cave were subsequently identified, as was a single specimen from the Baishiya Karst Cave on the Tibetan Plateau. DNA evidence suggests they had dark skin, eyes, and hair, and had a Neanderthal-like build and facial features. However, they had larger molars which are reminiscent of Middle to Late Pleistocene archaic humans and australopithecines.

18. Archaeopteryx is –

- (a) Most ancient bird of jurassic age
(b) Reptile of Jurassic period
(c) Reptile of Triassic period
(d) Reptile of both Triassic and Jurassic period

40th B.P.S.C. (Pre) 1995

Ans. (a)

Archaeopteryx is a connecting link between reptilia and aves. It is a creature in which we found the character of both reptiles and birds. Archaeopteryx dating from about 150 million years ago during the Jurassic period when many dinosaurs lived. It is one of the oldest-known birds. It is the first bird and modified from reptiles.

19. 'Archaeopteryx' is a connecting link between which of the following classes of animal?

- (a) Amphibia and Aves (b) Reptilia and Aves
(c) Reptilia and Mammalia (d) Aves and Mammalia

R.A.S./R.T.S. (Pre) 2008

Ans. (b)

See the explanation of above question.